

JUnit Testing Exercises

Solution:

Exercise 1: Create a Java Class to Test

Create a class named Calculator.java

```
public class Calculator {  
    public int add(int a, int b) {  
        return a + b;  
    }  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
    public int divide(int a, int b) {  
        return a / b;  
    }  
}
```

Exercise 2: Basic JUnit Tests

```
import static org.junit.Assert.*;  
import org.junit.Test;  
public class CalculatorTest {  
    Calculator calc = new Calculator();  
    @Test  
    public void testAdd() {  
        assertEquals(15, calc.add(10, 5));  
    }  
    @Test  
    public void testSubtract() {  
        assertEquals(5, calc.subtract(10, 5));  
    }  
    @Test
```

```

    public void testMultiply() {
        assertEquals(50, calc.multiply(10, 5));
    }
    @Test
    public void testDivide() {
        assertEquals(2, calc.divide(10, 5));
    }
}

```

Exercise 3: Assertions in JUnit

Create AssertionsTest.java

```

import static org.junit.Assert.*;
import org.junit.Test;
public class AssertionsTest {
    @Test
    public void testAssertions() {
        // Assert Equals
        assertEquals(5, 2 + 3);
        // Assert True
        assertTrue(5 > 3);
        // Assert False
        assertFalse(5 < 3);
        // Assert Null
        assertNull(null);
        // Assert Not Null
        assertNotNull(new Object());
    }
}

```

Exercise 4: AAA Pattern, Setup and Teardown

```

import static org.junit.Assert.*;
import org.junit.After;
import org.junit.Before;
import org.junit.Test;

```

```
public class CalculatorAAATest {
    Calculator calculator;
    // Setup method
    @Before
    public void setUp() {
        System.out.println("Setup executed");
        calculator = new Calculator();
    }
    // Teardown method
    @After
    public void tearDown() {
        System.out.println("Teardown executed");
        calculator = null;
    }
    @Test
    public void testAddition() {
        // Arrange
        int num1 = 20;
        int num2 = 10;
        // Act
        int result = calculator.add(num1, num2);
        // Assert
        assertEquals(30, result);
    }
    @Test
    public void testSubtraction() {
        // Arrange
        int num1 = 20;
        int num2 = 10;
        // Act
        int result = calculator.subtract(num1, num2);
        // Assert
        assertEquals(10, result);
    }
}
```

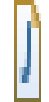
v CTS_EXERCISES

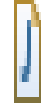
>  JRE System Library [JavaSE-22]

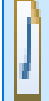
>  JUnit 4

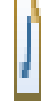
v  src

v  MODULE_1

>  AssertionsTest.java

>  Calculator.java

>  CalculatorAAATest.java

>  CalculatorTest.java

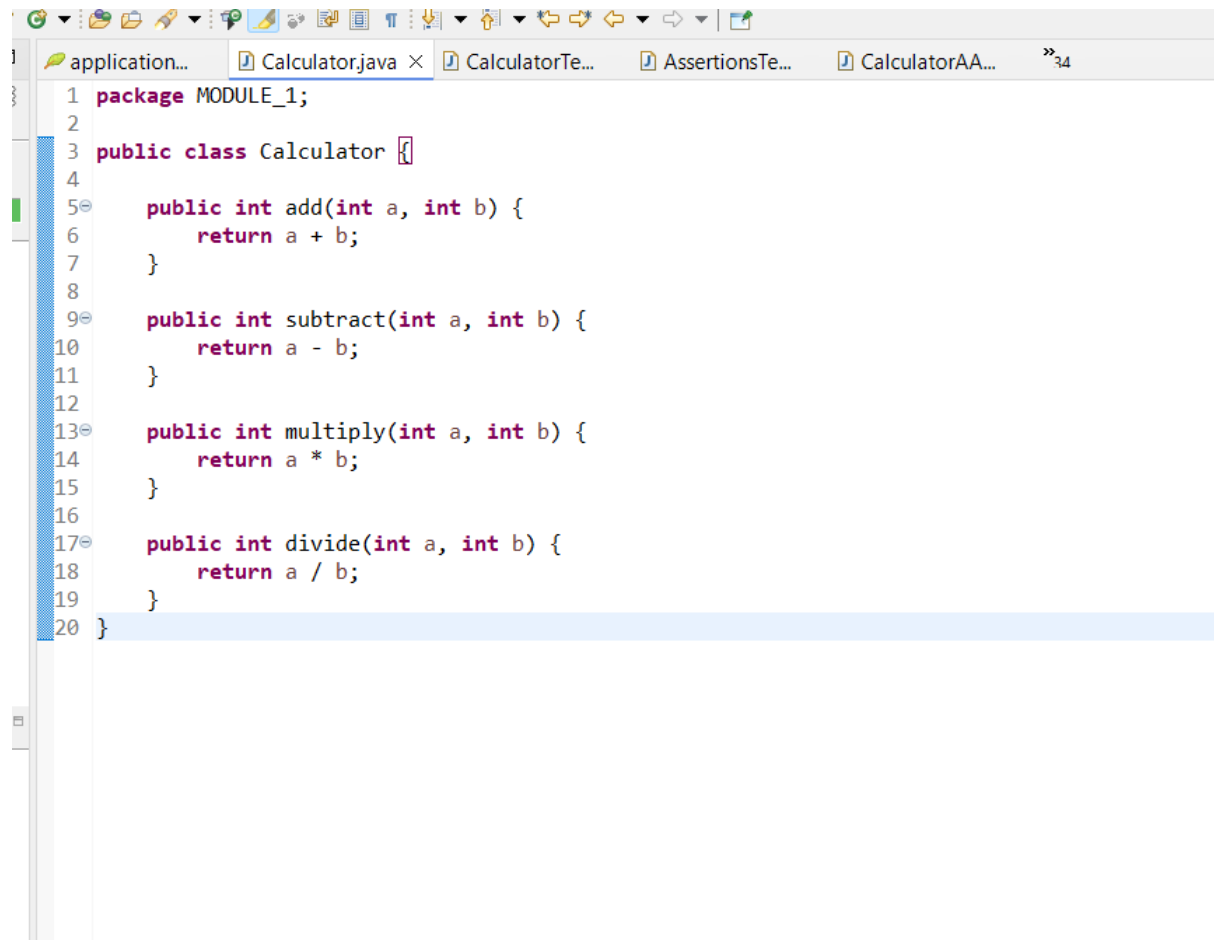
```
application... Calculator.java CalculatorTe... AssertionsTe... CalculatorAA... "34
1 package MODULE_1;
2
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5
6 public class CalculatorTest {
7
8     Calculator calc = new Calculator();
9
10    @Test
11    public void testAdd() {
12        assertEquals(15, calc.add(10, 5));
13    }
14
15    @Test
16    public void testSubtract() {
17        assertEquals(5, calc.subtract(10, 5));
18    }
19
20    @Test
21    public void testMultiply() {
22        assertEquals(50, calc.multiply(10, 5));
23    }
24
25    @Test
26    public void testDivide() {
27        assertEquals(2, calc.divide(10, 5));
28    }
29 }
30
```

Problems @ Javadoc Declaration Console x

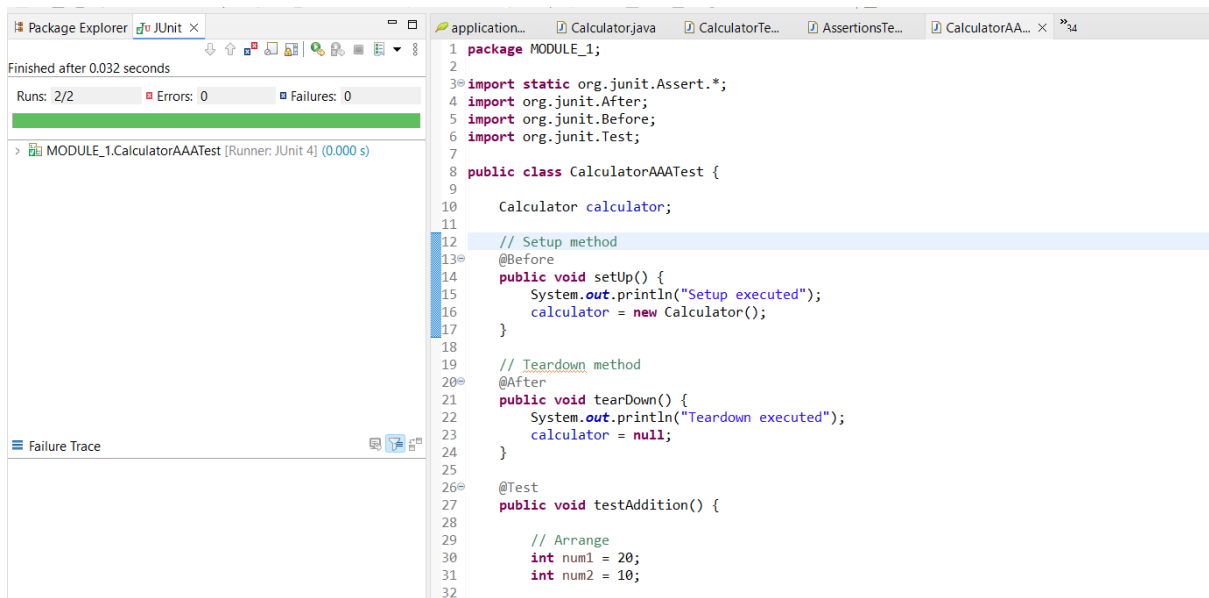
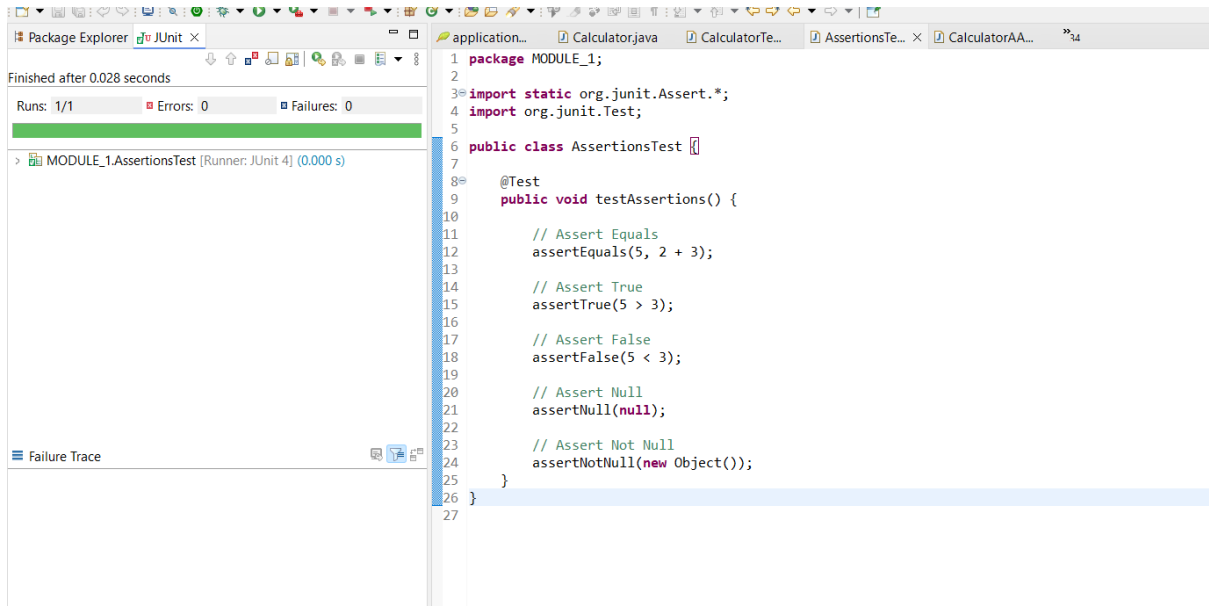
<terminated> CalculatorAAATest [I] Init! C:\Users\Sai Deenika\p2\pool\plugins\org.eclipse.jdt.openide.hotspot.ire.full.win32.x86_64.23.0.1.v2

application... Calculator.java CalculatorTe... AssertionsTe... × CalculatorAA... 34

```
1 package MODULE_1;
2
3 import static org.junit.Assert.*;
4 import org.junit.Test;
5
6 public class AssertionsTest {
7
8     @Test
9     public void testAssertions() {
10
11         // Assert Equals
12         assertEquals(5, 2 + 3);
13
14         // Assert True
15         assertTrue(5 > 3);
16
17         // Assert False
18         assertFalse(5 < 3);
19
20         // Assert Null
21         assertNull(null);
22
23         // Assert Not Null
24         assertNotNull(new Object());
25     }
26 }
27
```



```
1 package MODULE_1;
2
3 public class Calculator {
4
5     public int add(int a, int b) {
6         return a + b;
7     }
8
9     public int subtract(int a, int b) {
10        return a - b;
11    }
12
13    public int multiply(int a, int b) {
14        return a * b;
15    }
16
17    public int divide(int a, int b) {
18        return a / b;
19    }
20 }
```



Package Explorer JUnit x

CalculatorTest

Runs: 4/4 Errors: 0 Failures: 0

MODULE_1.CalculatorTest [Runner: JUnit 4] (0.004 s)

- testAdd (0.003 s)
- testSubtract (0.000 s)
- testDivide (0.000 s)
- testMultiply (0.001 s)